

Chapter - 4

Sources of working Capital :-

1) loans from financial institutions.

2) Floating of debentures.

3) Accepting public deposits.

4) Funds through internet.

Estimation of working Capital :

Prepare an estimate of working Capital requirement from the following information of a Trading concern.

Project annual sales 120000 units.

Selling price ₹ 10 per unit.

Percentage net profit as sells 30%.

Avg Credit period allowed to Customers 10 weeks.

Avg Credit period allowed to Suppliers 5 weeks.

Avg stock holding in terms of sales requirement weeks.

Allow 15% on Contingency.

Debtors = Customers / Credit Sales.

Creditors = Suppliers / Credit purchases.

Cost of Sales = Sales - Profit = 1200000 - 360000
= 840000

Sales = 120000 x 10 = 1200000

Profit = 1200000 x 30% = 360000

Estimation of working Capital.

Debtors = 840000 x 10/52

161538

Stock = 840000 x 5/52

80769

Total Current Asset

242307

less: Current liabilities (li) 840000 x 5/52

80769

161538

Add: Contingency 161538 x 15/100

24230

Net working Capital.

185768

2. Prepare An estimate of working Capital requirement from the following data of a trading concern.
- Projected annual sales 80000 units.
 - Selling price ₹ 8 / unit.
 - % net Profit as sales 20%
 - Avg Credit period allowed to Customers 10 week
 - " " Suppliers 8 week.
 - Avg Stock holding in terms of sales requirements 10 week
 - allow 20% Contingency.

$$\Rightarrow \text{Sales} = 80000 \times 8 = 640000$$

$$\text{Profit} = 640000 \times 20/100 = 128000$$

$$\text{Cost of Sales} = \text{Sales} - \text{Profit} = 640000 - 128000 = 512000$$

Estimation of Working Capital.

$$\text{Debtors} = 512000 \times 10/52$$

98461

$$\text{Stock} = 512000 \times 10/52$$

98461

Total Current Asset

196922

less: Current Liabilities.

$$\text{Creditors} = 512000 \times 8/52$$

78769

$$\text{Add: Contingency } 20\% \times 118152$$

118152

23630

Net working Capital

141782

Genaneshree Paints Ltd sells its product on a credit period of 20% on sales. The following information is extracted from annual accounts.

Sales at 3 months credit	4000000
Raw materials +	1200000
Wages paid 15 days in advance	960000
Manufacturing exp 1 month in advance	1200000
Administration exp 1 month in advance	480000
Sales Promotion exp 1/2 year in advance	200000
Income tax (Payable quarterly last instalment allowed due in December)	400000

The Company enjoys 1 month credit from supplies of raw materials & maintain 2 months stock of Raw materials and 1 1/2 month ^{stock} finished goods. Cash balance is maintain of 100000 has Precautionary balance. Assume 10% margin find out net working capital.

Estimation of working Capital

Debtors $(4000000 \times 3/12)$	1000000
Raw materials $(1200000 \times 2/12)$	200000
Finished goods $(3200000 \times 1.5/12)$	400000
Advance Sales $(200000 \times 1/12)$	100000
Cash	100000
Total Current Assets.	1800000
Less: C.L. wages paid. $(960000 \times 0.5/12)$	40000
Manuf exp $(1200000 \times 1/12)$	100000
Admin exp $(480000 \times 1/12)$	40000
Income tax $(400000 \times 1/4)$	100000
Creditors $(4000000 \times 1/12)$	333333
	380000
	1420000
Add: Contingency 10% X 1420000	142000
Net working Capital	1562000

Genaneshree Paints Ltd sells its product on a credit period of 20% on sales. The following information is extracted from annual accounts.

Sales at 3 months Credit

Raw materials 4000000

Wages paid 15 days in arrears 1200000

Manufacturing exp 1 month in arrears 960000

Administration exp 1 month in arrears 1200000

Sales Promotion exp 1/2 year in advance 480000

Income tax (Payable quarterly last instalment 200000)

(allows due in December) 400000

The Company enjoys 1 month Credit from suppliers of raw materials & maintain 2 months stock of

Raw materials and 1 1/2 month ^{stock} finished goods

Cash balance is maintain of 100000 has

Provisionary balance. Assume 10% margin

Find out net working capital

Estimation of working capital

Debtors (4000000 x 3/12) 1000000

Raw materials (1200000 x 2/12) 200000

Finished goods (3200000 x 1.5/12) 400000

Advance Sales (200000 x 1/12) 100000

Cash 100000

Total Current Assets 1800000

CL. wages paid. (960000 x 0.5/12) 40000

Manuf exp (1200000 x 1/12) 100000

Admin exp (480000 x 1/12) 40000

Income tax (400000 x 1/4) 100000

Debtors (4000000 x 1/12) 333333

1200000 380000

Contingency 10% x 1420000 142000

net working capital 142000

4

the Performance Cost sheet of a Company.

elements of Cost Amount per unit

materials 50%

labour 15%

overheads 15%

The following Particulars are available

It is proposed to maintain level of activity of 300000 units

Selling price is ₹ 20 per unit.

Raw materials are expected to be in stores avg of 2 months
materials will be in process avg of 1 month.

Finished goods are required to be in stock avg of 2 months.

Credit allowed to debtors 2 months.

Credit allowed by suppliers 2 months.

We may assume sales and production follow consistent pattern.

⇒ Current Asset

Raw materials $300000 \times \frac{50}{100} \times 20 \times \frac{2}{12} = 500000$

labour $300000 \times \frac{15}{100} \times 20 = 900000$

overheads $300000 \times \frac{15}{100} \times 20 = 900000$

Working Process (1 month)

Raw materials $300000 \times 20 \times \frac{50}{100} \times \frac{1}{12} = 250000$

labour $300000 \times 20 \times \frac{15}{100} \times \frac{1}{12} = 75000$

overhead $300000 \times 20 \times \frac{15}{100} \times \frac{1}{12} = 75000$

Finished goods (2 months)

Raw materials $300000 \times 20 \times \frac{50}{100} \times \frac{2}{12} = 500000$

labour $300000 \times 20 \times \frac{15}{100} \times \frac{2}{12} = 150000$

overhead $300000 \times 20 \times \frac{15}{100} \times \frac{2}{12} = 150000$

Debtors (2 month)

Raw materials	$300000 \times 20 \times \frac{50}{100} \times \frac{2}{12}$	500000
labour	$300000 \times 20 \times \frac{15}{100} \times \frac{2}{12}$	150000
overhead	$300000 \times 20 \times \frac{15}{100} \times \frac{2}{12}$	150000

Total Current asset

2500000

Less: Current liabilities. (2 months)

Raw materials	$300000 \times 20 \times \frac{50}{100} \times \frac{2}{12}$	500000
labour	$300000 \times 20 \times \frac{15}{100} \times \frac{2}{12}$	150000
overhead	$300000 \times 20 \times \frac{15}{100} \times \frac{2}{12}$	150000
		800000
Net working Capital		1700000

Statement of a Company provides following particulars.

material	40%
labour	20%
overheads	20%

The following particulars are available.

- It is proposed to maintain level of stock 200000 units.
- Raw materials are expected to remain in store avg of 1 month.
- Materials will be in process avg of $\frac{1}{2}$ month.
- Selling price of 12 per unit.
- Finished goods are required to be in stock avg of 1 month.
- Credit allowed to debtors 2 month.
- Credit allowed by supplier 1 month.

=>

Current Assets

Raw materials $200000 \times 12 \times 40/100 \times 1/12$ 80000

Working progress

Raw materials $200000 \times 12 \times 40/100 \times 0.5/12$ 40000

labour $200000 \times 12 \times 20/100 \times 0.5/12$ 20000

overhead $200000 \times 12 \times 20/100 \times 0.5/12$ 20000

Finished goods

Raw materials $200000 \times 12 \times 40/100 \times 1/12$ 80000

labour $200000 \times 12 \times 20/100 \times 1/12$ 40000

overhead $200000 \times 12 \times 20/100 \times 1/12$ 40000

Debtors

Raw materials $200000 \times 12 \times 40/100 \times 2/12$ 160000

labour $200000 \times 12 \times 20/100 \times 2/12$ 80000

overhead $200000 \times 12 \times 20/100 \times 2/12$ 80000

Total Current Asset 640000

less: Current liabilities (1 month)

Creditors:-

Raw materials $200000 \times 40/100 \times 12 \times 1/12$ 80000

labour $200000 \times 20/100 \times 12 \times 1/12$ 40000

overhead $200000 \times 20/100 \times 12 \times 1/12$ 40000 160000

Net working Capital.

480000

Vinay engineering Ltd request due to prepare a short showing working Capital requirements for a level of activity & 156000 units of Production. The following information for available for calculation

Particulars	Cost per unit (₹)
Raw materials	90
Direct labour	40
Overheads	75
Total Cost	205
Profit	60
Selling price	265

Raw materials are in stock as avg for 1 month
materials are in process (50%) an avg for 4 weeks.
Finished goods are in stock an avg for 1 month.
Credit allowed by Suppliers 1 month.

Time taking payment from debtors 2 months.

Avg lagging Payment of wages 1.5 weeks.

Avg lagging Payment of overheads 1 month.

20% of the O/P is sold against cash. Cash in hand are expected to be 60000. The production covered through out the year wages & overhead are similarly time period of 4 weeks is equivalent to a 1 month.

Current Asset

Raw materials.	$156000 \times 90 \times 4/52$	1080000
Working progress (4 weeks)		
Raw materials.	$156000 \times 90 \times 2/52$	1080000 540000
labour	$156000 \times 40 \times 2/52$	480000 240000
overhead	$156000 \times 75 \times 2/52$	400000 450000

+ Contingency 15% X 1780770

267115

Total working Capital.

2047885.5

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Date

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Following annual figure X relate to X company.

Sales (2 months credit) 3600000

Materials Consumed (Suppliers extent 2 months credit) 900000

wages paid (monthly in arrears) 720000.

Manufacturing expenses etc at the end of year 80000.

(The expenses paid 1 month arrears)

Total administration expenses paid has above 240000.

Sales promotion expenses paid quarterly in advance 120000.

The Company sells its products at a gross profit 25%

Counting depreciation the cost of production.

It keeps 1 month stock of raw materials & finished goods

& Cash Balance ₹ 100000.

Assume 20% Safety margin Calculate working Capital requirements.

⇒ Cost of Sales.

Sales 3600000

- gross profit 25% 900000

Cost of Sales 2700000

Current asset:

Raw materials $(2700000 \times \frac{1}{12}) (900000 \times \frac{1}{2})$ 225000 75000

Finished goods $(2700000 \times \frac{1}{12}) (2700000 \times \frac{1}{12})$ 225000 225000

Debt $(2700000 \times \frac{2}{12}) (3600000 \times \frac{2}{12})$ 450000 600000

Sales promotion $(120000 \times \frac{1}{4}) (120000 \times \frac{1}{4})$ 30000 30000

Cash 100000 100000

Total Current asset 1030000

less: Current liabilities.

Credits. $(900000 \times \frac{2}{12})$ 150000

wages. $(720000 \times \frac{1}{12})$ 60000

Manufacturing exp $(80000 \times \frac{1}{12})$ 6667

Net working Capital. 793333
 Add: Contingency 20% 158667
 Total working Capital 952000

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 Date 23/8/23
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You are given the following estimates and are interested to compare with Capital.

Particular	Amount
Stock of finished goods ₹ 50	3000
Stock of materials & stores	8000
Avg Credit given :-	
Domestic Sales 6 weeks Credit	312000
Exports Sales 10.5 weeks Credit	78000
Avg time lagging payment :-	
wages 10.5 weeks	260000
materials 10.5 months	48000
Rent 6 months	10000
clerical staff 0.5 months	62400
Manager 0.5 month	4800
other expenses 10.5 month	48000
Sundry expenses Credit quarterly in advance	8000

The Stmt showing details of working Capital

Current Asset :-		
finished goods		5000
Raw materials		8000
D/S.		
Domestic Sales	$312000 \times 6/52$	36000
Export Sales	$78000 \times 10.5/52$	2250
Sundry expenses	$8000 \times 1/4$	2000
Total Current asset		53250
Current Liabilities :-		
wages	$260000 \times 10.5/52$	7500
materials	$48000 \times 10.5/12$	6000
Rent	$10000 \times 6/12$	5000
clerical staff	$62400 \times 0.5/12$	2600
Manager	$4800 \times 0.5/12$	200
other expenses	$48000 \times 10.5/12$	22200

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Estimate the working capital requirement from the following information you are given the following estimate & are instructed to add 20% Contingency.

Particulars	Amt (₹)
Stock of finished goods	6000
Stock of material	10000
Avg Credit Sales %	
England Sales 8 weeks credit	300000
Export Sales 2 weeks credit	80000
Lagging payment of wages & other outgoing %	
wages 2 weeks	250000
Stock of materials 2 months	50000
Rent Royalties 6 months	10000
Classical staff 1 month	5000
Miscellaneous expenses 2 months	50000
Payment in advance %	
Sundry expenses (Quarterly)	10000

⇒

Current Asset :

Stock of finished goods

Stock of material

Debt :England Sales $(300000 \times \frac{8}{52})$ Export Sales $(80000 \times \frac{2}{52})$ Sundry expenss. $(10000 \times \frac{1}{4})$

Total Current Asset →

less: Current Liabilities :wages $(250000 \times \frac{2}{52})$ Stock of materials $(50000 \times \frac{2}{12})$ Rent $(10000 \times \frac{6}{12})$ Classical staff $(5000 \times \frac{1}{12})$ Miscellaneous $(50000 \times \frac{2}{12})$

6000

10000

46154

3077

2500

67731

9615

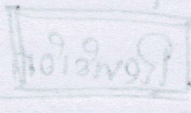
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5000

417

8323

31698



the following projection of XYZ Ltd for the next year
you are required to determine working capital.

Annual sales 1440000
Cost of production (including depreciation) 120000 120000

Raw materials purchase 705000

Monthly expenditure 30000

Estimated opening stock of Raw material 140000

Estimated closing stock of Raw material 125000

Inventory norms:-

Raw materials 2 months, working progress 1/2 month &
finished goods 1 month.

The firm enjoys a credit of 1/2 month on its purchase
and allow 1 month credit on its supplies.

On sales order the company receive advance 15000.

He may resume production covered out throughout the yr.

And minimum cash balance of 35000.

Revision :

Total Cost & level of activity.

The performance cost sheet of a Company provide the following particulars:

elements of Cost	Amt / unit
material	₹ 80
labour	₹ 30
Overhead	₹ 60
Total Cost	170
Profit	30

Selling price

The following further particulars are available.

Raw materials are in stock an average for 1 month

Raw materials are in process an average for $\frac{1}{2}$ month

finished goods are in stock on an avg for 1 month

Credit allowed by Suppliers 1 month

lagging Payment of wages $1\frac{1}{2}$ week

lagging Payment of overhead 1 month

$\frac{1}{4}$ th output is sold against cash

Cash in hand & Bank expected to be ₹ 25000

Credit allowed to Customers 2 months.

you are required to prepare a stmt showing working capital & level of activity 104000 units of prod

Production carried through out 4 year wages & overhead accrued time period of 4 weeks equivalent to 8 month.

$\frac{1}{12}$

Solution

Current Assets :

Raw materials. $104000 \times \frac{1}{12} \times 80$ 640000.

working progress :

Raw materials $104000 \times \frac{1}{12} \times 80$ 320000

labour $104000 \times 30 \times \frac{1}{12}$ 120000

overhead $104000 \times 60 \times \frac{1}{12}$ 240000

Finished goods:

Raw materials	$104000 \times 80 \times \frac{4}{52}$	640000
labour	$104000 \times 30 \times \frac{4}{52}$	240000
overhead	$104000 \times 60 \times \frac{4}{52}$	450000

Debtors:

Raw materials	$104000 \times 80 \times \frac{8}{52} \times \frac{3}{4}$	960000
labour	$104000 \times 30 \times \frac{8}{52} \times \frac{3}{4}$	360000
overhead	$104000 \times 60 \times \frac{8}{52} \times \frac{3}{4}$	720000
Cash		25000
Cash in hand		474000
Total Asset		474000

Liabilities:

Raw materials	$104000 \times 80 \times \frac{4}{52}$	640000
labour / wages	$104000 \times 30 \times \frac{4}{52}$	90000
overhead	$104000 \times 60 \times \frac{4}{52}$	420000
Net working Capital		1210000
		3535000

Vinay engineering private limited request you to prepare statement working capital requirement level of activity 156000 units.

Raw material	90
labour	40
overheads	75
Total Cost	205
Profit	60
Selling price	265

Raw materials are in stock avg of 1 month
 " are in process (50% completed) as avg 4 week.
 Finished goods " stock an avg of 1 month
 Credit allowed by supplier 1 month
 Average lagging Payment from debtor 2 month
 Avg " of wages 15 weeks
 " of overhead 1 month.